

HAMMARLAND MARKET, FinlandWMO: **029930**Lat: **60.301N**Lon: **19.131E**Elev: **50**StdP: **14.67**Time Zone: **2.00 (EUE)**Period: **04-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	15.1	19.8	8.0	8.3	17.4	12.1	10.2	20.6	44.9	32.9	40.4	31.7	18.2	100	0.893

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.4	71.0	67.3	68.7	64.9	66.6	62.8	68.5	70.8	65.7	68.1	63.9	66.2	14.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	101.2	69.9	64.8	91.9	67.3	62.8	85.7	65.5	32.8	70.9	30.6	68.2	29.2	66.5	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 40.4	35.6	32.1	DB	14.0	73.2	7.6	3.2	8.5	75.6	4.0	77.4	-0.2	79.2	-5.7	81.6	(4)
(5)			WB	11.8	67.8	7.4	3.2	6.5	70.2	2.2	72.0	-2.0	73.9	-7.3	76.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.5	31.6	29.9	32.7	37.9	45.0	53.2	61.4	62.1	56.0	47.1	40.1	36.1	
	DBStd	11.97	5.26	6.55	5.13	3.81	4.43	4.42	4.31	3.45	3.74	4.16	4.79	4.66	
	HDD50	3058	571	562	538	364	165	19	0	0	3	108	296	432	
	HDD65	7505	1036	982	1003	814	619	353	129	103	269	554	746	897	
	CDD50	1060	0	0	0	0	11	116	354	375	185	19	0	0	
	CDD65	30	0	0	0	0	0	0	18	12	0	0	0	0	
	CDH74	10	0	0	0	0	0	0	10	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	17.3	21.0	18.6	16.5	15.2	15.0	14.7	13.8	15.2	16.8	19.0	20.4	22.0	
Precipitation	PrecAvg	21.5	1.7	1.3	0.9	1.1	1.5	2.0	1.9	2.4	2.2	2.3	2.2	1.9	
	PrecMax	30.9	3.4	3.1	2.4	2.6	3.4	3.7	5.2	6.6	4.8	5.6	5.6	3.8	
	PrecMin	16.4	0.0	0.1	0.0	0.2	0.1	0.5	0.2	0.1	0.3	0.0	0.4	0.2	
	PrecStd	3.2	0.9	0.7	0.6	0.7	0.9	0.9	1.5	1.5	1.2	1.3	1.1	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.4	42.2	45.7	51.0	58.4	65.4	75.1	72.4	65.6	57.8	50.2	48.3	
		MCWB	40.7	38.4	40.6	44.2	50.6	57.4	68.1	68.7	61.4	55.3	48.5	45.7	
	2%	DB	41.1	39.6	42.4	47.1	55.0	62.9	72.2	70.4	63.8	55.6	48.6	44.9	
		MCWB	39.3	36.8	38.4	42.2	49.3	57.3	68.2	66.5	60.4	53.3	47.1	42.3	
	5%	DB	39.8	38.2	40.3	44.7	52.7	61.0	69.9	68.4	62.3	54.0	47.5	43.4	
		MCWB	38.3	36.0	37.2	40.7	48.4	56.3	66.5	64.4	59.2	52.0	46.0	41.2	
	10%	DB	38.5	36.7	38.5	42.7	50.7	59.1	67.5	66.6	60.9	52.8	46.1	41.9	
		MCWB	37.0	35.1	36.2	39.5	47.4	55.3	64.1	62.9	57.9	50.7	44.7	40.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.4	39.4	41.4	45.4	53.1	60.5	71.3	71.0	63.4	56.2	49.2	46.2
			MCDB	42.5	41.3	45.1	49.5	57.2	63.2	73.0	71.6	64.3	57.1	49.6	48.2
2%		WB	39.8	37.6	39.2	42.9	50.4	58.5	69.2	67.9	61.2	53.9	47.6	43.2	
		MCDB	40.7	39.3	42.0	45.8	53.2	61.3	71.4	70.0	63.2	55.0	48.4	44.5	
5%		WB	38.6	36.5	37.7	41.3	49.1	57.2	67.3	65.3	59.8	52.7	46.4	41.7	
		MCDB	39.6	37.9	39.7	43.9	51.9	59.8	70.0	67.4	61.8	53.9	47.4	43.0	
10%		WB	37.2	35.5	36.6	40.0	47.7	55.8	64.6	64.0	58.5	51.3	45.0	40.3	
		MCDB	38.4	36.7	38.4	42.3	50.3	58.4	67.4	66.2	60.4	52.7	46.1	41.8	
Mean Daily Temperature Range	5% DB	MDBR	4.7	4.6	4.7	5.3	5.7	5.6	4.5	4.4	4.4	4.0	4.2	4.7	
		MCDBR	5.2	5.3	7.1	9.0	9.3	8.6	6.7	5.8	5.5	4.7	4.4	5.7	
		MCWBR	5.1	4.5	5.2	5.4	5.4	5.1	4.9	4.5	4.3	4.7	4.4	5.4	
	5% WB	MCDBR	5.1	5.0	6.5	8.7	8.5	7.4	5.7	4.4	4.6	4.7	4.2	5.3	
		MCWBR	5.0	4.3	5.0	5.8	5.6	5.0	5.0	4.3	4.1	4.9	4.4	5.4	
Clear-Sky Solar Irradiance	taub	0.274	0.313	0.332	0.368	0.367	0.358	0.385	0.375	0.345	0.331	0.297	0.382		
	taud	2.199	2.277	2.341	2.343	2.419	2.487	2.432	2.450	2.520	2.421	2.230	2.654		
	Ebn at Noon	155	215	252	263	272	277	265	259	249	209	142	109		
	Edh at Noon	13	21	27	33	33	31	32	29	23	18	12	9		
All-Sky Solar Radiation	RadAvg	76	253	732	1256	1702	1905	1775	1380	856	367	104	49		
	RadStd	9	50	80	109	138	125	172	120	88	57	24	7		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-191	N/A	N/A	N/A

Nomenclature: See separate page